INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 2: Literature Review and Theoretical Framework

by

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2 Literature Review and Theoretical Framework

The research problem set out in Chapter 1 is that transfer-graph reputation scores give no role to spending authorization, an explicit act of trust that every token following the ERC-20 standard records, and that their graphs grow with every payment. The objectives O1–O5 and research questions RQ1–RQ5 follow from it. The addresses ranked are smart contracts: the routers, vaults, aggregators, lending pools and other contracts that owners approve to spend their tokens. To compare EndorseRank with Adaptive Weighted PageRank (AWP) on these contracts, we run each method's graph through PageRank and check the resulting rankings with rank correlations. The framework combines hyperlink–citation propagation, domain alignment and computational complexity. The gap that remains is taken up in Chapters 3 and 4.

2.1 Blockchain and DeFi foundations

A public blockchain records transactions in hashed blocks, and each block is secured by the ones added after it (Nakamoto, 2008). Once a majority of the participating nodes has accepted a block, rewriting it is costly. That cost allows parties to settle on the ledger without each of them needing a known legal identity (*trust-minimized* settlement).

Bitcoin built a peer-to-peer electronic cash network on this kind of ledger (Nakamoto, 2008). Ethereum keeps the append-only ledger but lets an account hold a program alongside its balance. A transaction can then call one of the program's functions and change the contract's state (Buterin, 2014), and these functions run on the Ethereum Virtual Machine (EVM) (Wood, 2025). Programs stored on the ledger in this way are called smart contracts, a term that Szabo (1997) used for contract terms embedded in computer protocols. Execution on the EVM is deterministic, so any honest node that replays a function call gets the same output (Wood, 2025). As a result,

collateralization ratios, liquidation conditions and fee schedules can be enforced by the contract instead of by a trusted third party (Schär, 2021). The contracts that users let act on their tokens are the objects this thesis ranks.

Collectively, these contracts make up Decentralized Finance (DeFi). Between them they do the work of exchanges, lending platforms, derivatives markets and asset managers; Werner et al. (2022) define the protocol classes, and Schär (2021) describes how protocols are layered on one another. Because one protocol can interact with another, new financial instruments can be composed from existing ones (Kitzler et al., 2023). In such compositions contracts call other contracts, and a contract can be the owner of an allowance as well as its spender, since the owner is simply the account that grants it (Vogelsteller & Buterin, 2015). The same property lets a failure spread across several protocols. Successful attacks have included flash-loan exploits (Qin et al., 2021) and price-oracle manipulation (Angeris & Chitra, 2020). Transaction histories on these chains are public, yet the addresses in them are pseudonymous, which makes the identity checks of conventional finance hard to carry out (D. Lin et al., 2025).

2.1.1 Pseudonymity and the credit problem

Conventional credit scoring ties a credit history to a legal entity. On a public blockchain the basic unit of analysis is instead an address. It may be an externally owned account controlled by a private key or a contract governed by its code, and it may stand for a legal entity, a trust, an algorithmic agent or a protocol. One entity can own several addresses (Douceur, 2002), and address-clustering heuristics can recover ownership only in part (Victor, 2020). On-chain lending therefore requires borrowers to post collateral worth more than the amount they borrow; if the collateral value falls below a set threshold, a liquidator may repay part of the debt and receive the corresponding collateral at a discount (Bartoletti et al., 2021). The arrangement is secure, but it ties up capital. An on-chain credit score tries to quantify from on-chain behavior alone how likely an actor is to repay. Such scores draw on prospect theory (Hassija et al., 2020) and on machine learning applied to lending data (Kotb, 2023), and a related line of work rates the fraud risk of Ethereum accounts by propagat-

ing it through the transaction network (D. Lin et al., 2025). More recent work predicts liquidations from DeFi transaction features gathered on several chains (Palaiokrassas et al., 2024), and a credit score for DeFi lending combines account-level financial features with relational features from a weighted transaction graph (M et al., 2026).

All of these scores rank borrowers. A contract does not borrow, but the same lack of identity arises when an owner decides whether to let a contract spend its tokens. The contract's code is public, while who deployed it, who can change it and how it will use the allowance may not be. A reputation score for contracts asks what the on-chain record of other owners' decisions says about a given contract.

2.2 Ethereum, Layer-2 rollups, and Arbitrum One

Most DeFi applications were first deployed on the Ethereum mainnet. While Ethereum ran on proof of work, it could not raise its block size or shorten its block interval far without weakening its security (Gervais et al., 2016), so growing use of the mainnet raised gas costs. Layer-2 (L2) protocols try to keep Ethereum's security guarantees while making transactions cheaper. Some execute transactions off-chain, while others batch them and post a succinct summary of the execution to the Ethereum mainnet (Gudgeon, Moreno-Sanchez, et al., 2020). Thibault et al. (2022) survey the protocols of the second kind, called rollups, and compare their implementations. Optimistic L2s such as Arbitrum treat each submitted batch of transactions as valid unless evidence against it is presented within a dispute period (Kalodner et al., 2018).

Arbitrum One is an optimistic L2 of this kind and can execute EVM-based smart contracts. It has become a popular platform for DeFi applications, among them GMX (n.d.), an exchange for perpetual swaps. Contracts on Arbitrum do not simply behave as they would on Ethereum mainnet: a study of contracts migrated from Ethereum to Arbitrum traces their security risks to differences in cross-chain messaging, block properties, contract address aliasing and gas fees (Tang & Shi, 2024). We use Arbitrum One because the public BigQuery dataset includes both its ERC-20 `Approval`

and `Transfer` events, and because its fees are low enough that the observation window contains dense activity around the contracts studied. We make no claim that our results generalize to other L2s or to Ethereum mainnet without further study. The framework (approval and transfer graphs, construct checks and rank correlation) can, however, be applied to any chain that emits the same logs.

2.3 ERC-20 tokens and the allowance lifecycle

The ERC-20 standard defines a common interface for fungible tokens, made up of `transfer`, `approve`, `transferFrom`, `allowance` and the associated events (Vogelsteller & Buterin, 2015). After the owner calls `approve(spender,amount)`, the token contract records that the spender may move up to `amount` out of the owner's balance. The spender can then call `transferFrom` to carry out a token transfer. With EIP-2612 `permit`, an allowance can be updated through an off-chain signature that settles in a single transaction (Lundfall, 2020). The user experience improves, and the underlying semantics of authorization are unchanged. The owner is whichever account grants the allowance, an EOA or a contract, and the spender can be any address; in this thesis the spenders that are scored are contracts.

Figure 2.1 traces an ERC-20 allowance through its lifecycle. An allowance is an authorization and involves no actual transfer. Token owners can grant allowances to routers, vaults, aggregators, lending pools or trading contracts that they trust to act within set limits. Revoking (approving zero) and updating (setting a new non-zero allowance) both change the allowance currently in force. EndorseRank accounts for this by building each owner–spender edge from the most recent allowance on each of the five tokens it uses (Section 3.5.1), so a newer allowance replaces the previous one outright; old values are not averaged in or left to decay gradually.

Allowances are sparse compared with transfers. A single approval can cover many transfers, and many accounts never grant an approval to any other address. The sparsity lowers the cost of a PageRank solve, as a consequence of the edge count (Section 2.5.2), and it also marks a semantic

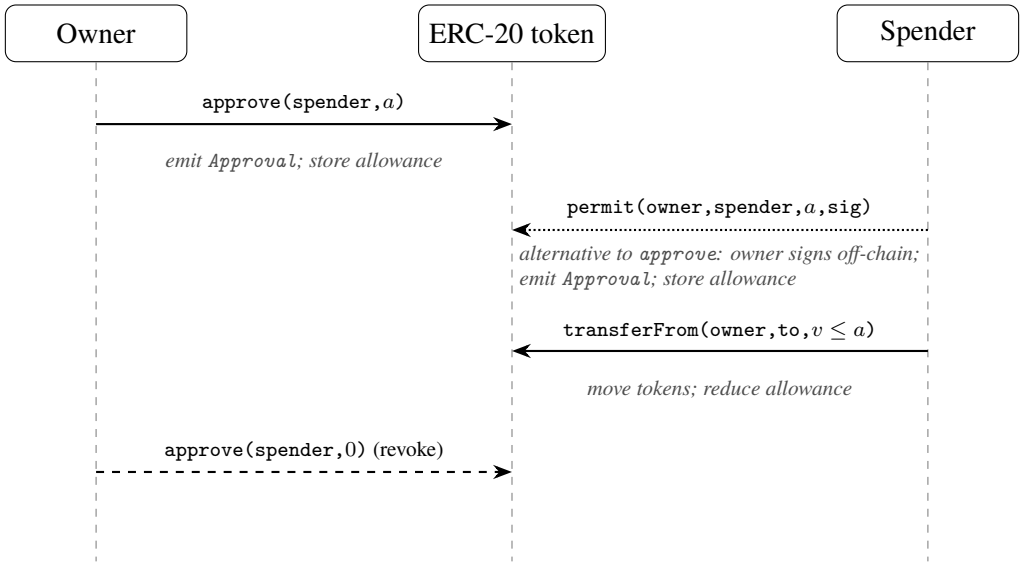


Figure 2.1: ERC-20 allowance lifecycle: `approve`, or its signature-based alternative `permit` (dotted), an optional `transferFrom`, and revoke. An EndorseRank edge sums, over the five tokens of the study, one term for the most recent `Approval` event of its owner–spender pair in the observation window, and the term weights that approval by its age and its US-dollar amount as AWP weights a transfer. A later approval replaces an earlier one.

difference, since EndorseRank quantifies authorization to spend whereas a transfer graph records payment flows.

2.4 Graph-based on-chain reputation

PageRank and its variants rank nodes by the influence that flows to them along directed edges (Page et al., 1999). Two recent studies apply this approach to blockchain transaction graphs, and the two should be distinguished. H.-D. Do and Do (2023) run PageRank and HodgeRank on Ethereum transactions as a measure of social credit, meant for people who lack a credit history at financial institutions. T. Do et al. (2023) propose Adaptive Weighted PageRank (AWP), the transfer-graph method that this thesis evaluates beside EndorseRank on the same contracts (Section 2.18). D. Lin et al. (2025) treat the risk rating of Ethereum accounts as a node-ranking problem and propagate risk through the transaction network, so that accounts without labels are rated as well, and Wu et al. (2022) detect phishing addresses from a network embedding of the transaction graph that takes the

amounts and times of transactions into account. Random walks have also been used to detect scam behavior on Ethereum from a graph of interactions between addresses (Yan et al., 2023), and a graph convolutional network labels the business role of Ethereum accounts from their features and their position in the transaction network (D. Lin et al., 2024). Bellini et al. (2020) survey blockchain-based trust and reputation management, and Almasoud et al. (2020) review reputation systems built on smart contracts, in which users rate how trustworthy the providers of online services or products are. Data mining on cryptocurrency transactions likewise works from transaction histories, for address linking and for the study of user behavior (X. F. Liu et al., 2021). Packin and Lev-Aretz (2024) warn that decentralized credit scores can turn biased decisions into automated processes that outsiders cannot inspect.

Outside the research literature, trust in deployed DeFi and web3 systems is usually established another way. An address builds standing through activity that anyone can inspect on the ledger, and tooling bound to the address summarizes it. One such tool is Human Passport, formerly Gitcoin Passport (Human Passport, n.d.). It collects verifiable credentials, called stamps, for checks such as government identity, biometrics, a web of trust and earlier web2 and web3 activity. Each stamp carries a weight that the documentation describes as the strength of its human signal, and an address passes when the sum of its weights reaches a threshold. According to the same documentation, Passport also classifies EVM addresses on several networks, Arbitrum among them, as human or Sybil from their on-chain activity. Its purpose is Sybil resistance, so it combines on-chain with off-chain evidence and works as a gate and not as a graded ordering. A review of proof-of-personhood protocols describes how they resist Sybil attacks with subjective inputs such as vouching and voting in place of attestations from central issuers (Siddarth et al., 2020), and a recent review of Sybil attacks on blockchains traces airdrop farming on layer-2 networks such as Arbitrum and assesses proof-of-personhood systems among the defenses (Bordeianu & Popescu, 2026).

Such tools answer whether an address is controlled by a distinct person. That is a question about the holder of a key, and it has no direct counterpart for a contract, whose behavior is fixed by its code and by whoever may upgrade it. Trust in a contract is formed in other ways. An owner usually

decides whether to trust a spender contract at the moment its wallet software presents the approval for signature, and approval phishing exploits that moment (Section 2.17); code analysis judges the contract by its functions (Section 2.14). EndorseRank answers a different question at the level of the contract: which contracts other addresses authorize to spend their tokens, and how strongly those addresses are endorsed in turn. It orders the contracts from on-chain records, with amounts valued at public daily token prices, and certifies nothing about the people behind them (Table 2.1). Graph-based methods need only public information, their scores are continuous and so can order every address, and their cost can be stated in terms of the edge count (Section 2.5.2). They are, however, vulnerable to Sybil attacks (Douceur, 2002) and wash trading, and graph centrality is neither the safety of a contract nor the default risk of an account; PageRank scores are not credit scores in the regulatory sense, and they are not security ratings (Section 2.8).

A propagated score must also beat a simpler one. Where degree correlations are weak, as on the Web graph, PageRank is on average proportional to in-degree and can be approximated from it (Fortunato et al., 2008). New links concentrate on nodes that already have many, as preferential attachment assumes (Barabási & Albert, 1999) and measured network growth shows (Newman, 2001), so link prediction uses degree-based predictors as standard baselines (Liben-Nowell & Kleinberg, 2007). On Ethereum, the frequency and the amount of past transactions both bear on which new transactions appear (D. Lin et al., 2022).

2.5 PageRank formalism

Consider a directed graph $G = (V, E)$ with $|V|$ nodes, and write $w_{ij} > 0$ for the weight of the edge from i to j. That weight stands for influence from i to j, in either the endorsement or the transfer sense introduced above. Let W be the corresponding weighted adjacency matrix, and let

$o_i = \sum_j w_{ij}$ be the out-strength of node i. The row-stochastic transition matrix P is then

$$P_{ij} = \begin{cases} w_{ij}/o_i & \text{if } o_i > 0, \\ 1/|V| & \text{if } o_i = 0 \end{cases} \tag{2.1}$$

(dangling nodes spread their mass evenly). Given a damping parameter $d \in (0, 1)$, the Google matrix is

$$G_{\text{PR}} = d\, P^\top + \frac{1-d}{|V|}\, \mathbf{1}\mathbf{1}^\top, \tag{2.2}$$

and the PageRank vector $\mathbf{r}$ satisfies $\mathbf{r} = G_{\text{PR}}\mathbf{r}$ with $\mathbf{r} \geq 0$ and $\sum_i r_i = 1$ (Langville & Meyer, 2006). Element by element, with the dangling rows of P written as a separate middle term, this becomes

$$r_j = d \sum_{i:(i\to j)\in E} r_i \frac{w_{ij}}{o_i} + \frac{d}{|V|} \sum_{i:\, o_i = 0} r_i + \frac{1-d}{|V|}. \tag{2.3}$$

2.5.1 Random-surfer interpretation

Equation (2.3) can be read probabilistically. At each step a surfer at a node with out-edges follows one of them with probability d, choosing it in proportion to its weight, or teleports to a uniformly chosen node with probability $1 - d$ (Figure 2.2). At a node without out-edges the surfer always jumps to a uniformly chosen node; this is the dangling rule of Equation (2.1). The stationary distribution then ranks nodes by how often they are visited in the long run. Damping keeps rank sinks (groups of nodes with no edge leaving them) from absorbing all the mass. With $d < 1$ and uniform teleportation, every entry of G_{PR} is positive, so G_{PR} is primitive and its Perron vector is unique (Langville & Meyer, 2006). Appendix 8.3.1 gives the argument.

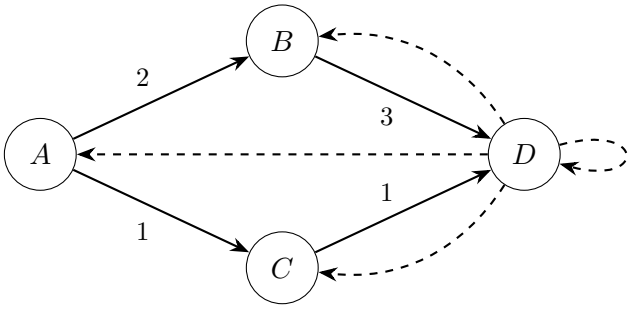


Figure 2.2: Random-surfer intuition for PageRank on the four-node graph of Section 2.13; edge labels are the allowance weights. Solid arrows are graph transitions: with total probability d the surfer follows one of them, chosen in proportion to its weight. Dashed arrows show uniform teleportation, which reaches every node (the current one included) with probability $(1-d)/|V|$; the dashed fan is drawn only from D, although every node has one. D has no outgoing edge, so under the dangling rule of Equation (2.1) a surfer at D always jumps to a uniformly chosen node, each with probability $1/|V| = 0.25$. That rule, and not damping, keeps D from absorbing the chain.

2.5.2 Power iteration and complexity

In practice, $\mathbf{r}$ is usually obtained by power iteration from the uniform vector $\mathbf{r}^{(0)} = \mathbf{1}/|V|$,

$$\mathbf{r}^{(t+1)} = G_{\text{PR}}\mathbf{r}^{(t)}, \tag{2.4}$$

until $\|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}\|_1 < \varepsilon$ or until the iteration cap $I_{\max}$ is reached. On a sparse graph, each iteration costs $O(|E|)$ arithmetic operations (Leskovec et al., 2020). A method that generates fewer edges, for instance by using latest-allowance snapshots instead of full transfer histories, therefore lowers the cost per iteration and sometimes the memory required. Hypothesis H3 is motivated by this complexity fact.

By default this work sets damping to the conventional $d = 0.85$ (Page et al., 1999). The convergence tolerance ε and the iteration cap $I_{\max}$ are study-specific solver settings, given with their values and the reason for them in Section 3.1.[1] The damping sensitivity checks of Section 4.5 vary d.

[1] They are not put forward as part of the classical PageRank formulation.

2.6 Reputation systems lineage

PageRank belongs to a wider family of spectral and propagation-based reputation mechanisms. The HITS algorithm of Kleinberg (1999) assigns each page two scores, one as a hub and one as an authority. EigenTrust extends the idea to peer-to-peer networks and normalizes local trust, so a malicious node cannot raise its own score at will (Kamvar et al., 2003). To suppress web spam, TrustRank starts from a seed set of trusted nodes and propagates their trust outward (Gyöngyi et al., 2004). Reputation is relational in all of these methods. Where trust flows only from a vetted seed set, as in TrustRank, a newcomer cannot raise its score without endorsements from nodes that already hold reputation. With uniform teleportation, which EndorseRank uses (Section 3.6.2), that protection is weaker. Gyöngyi and Garcia-Molina (2005) derive the link farms and farm alliances that most raise a target's PageRank, and Cheng and Friedman (2005) show that symmetric reputation functions such as PageRank can be manipulated with Sybils, whereas some asymmetric functions based on flows or paths from a trusted node resist them. The Sybil cost model of Section 5.6 prices such a farm on an allowance graph.

On chain, the Sybil attack (Douceur, 2002) is the main concern for a graph score. New addresses cost little to create, whereas addresses that receive allowances or transfers from distinct counterparties cost more. For a spender contract the cheap route is plain: whoever deploys it can create owner addresses and have each of them approve it, paying only gas. Coordinated multi-address behavior does leave traces in the transaction graph. Graph learning on NFT transactions identifies airdrop hunters who amass many addresses (C. Zhou et al., 2024), and features of each address's transaction subgraph, among them the timing of its first transaction and of its first gas funding, separate Sybil addresses in token airdrops (Q. Liu et al., 2025). The Sybil-adjusted stability proxies used in this work (inbound counterparty ratio, tenure, active months) partly capture that diversity of interaction, but they do not prevent a malicious actor from manipulating them, and no Sybil detector is applied here.

Machine-learning credit models fitted to DeFi lending data (Kotb, 2023) learn from labeled credit

outcomes, and an ensemble of fitted models is harder to audit than a fixed rule. EndorseRank and AWP apply fixed propagation rules that anyone can recompute from the public logs and a public series of daily token prices (Sections 2.4 and 3.5.1).

2.7 ERC-20 allowances as endorsement

Through the ERC-20 `approve` function (Vogelsteller & Buterin, 2015) and EIP-2612 `permit` (Lundfall, 2020), owners state explicitly how much a spender may move on their behalf. EndorseRank takes the latest allowance from an owner to a spender on each of the five tokens of the study (Section 3.5.1) as one endorsement, weights its US-dollar value as AWP weights a transfer, and runs PageRank on the resulting directed graph. Newer approvals supersede older ones, so each pair and token contributes a single event. The endorsed node is the spender contract, and the endorsing node is the owner, which may be an EOA or another contract.

Concretely, for each owner–spender pair the latest in-window approval on each of these tokens contributes its time weight times the bounded value weight of its US-dollar amount, as AWP weights a transfer; a token on which the pair's latest value is zero (a revocation) contributes no term, an unlimited allowance takes its full time weight, and the terms are summed over the tokens (Equation (3.7) in Section 3.6.2). The resulting graph G_{approve} is sparser than the transfer graph over the same contracts and window (Chapter 4 gives both edge counts), and PageRank on it yields the EndorseRank scores $\mathbf{s}_{\text{ER}}$ of the cohort contracts.

Treating allowances as endorsements is a deliberate choice, and an imprecise one. Owners approve routers because it is convenient, not necessarily to make a social statement about whether the router is trustworthy. Contract addresses and EOAs coexist on the owner side as well. A contract that approves another contract does so because its code says so, and the approval records a decision its developers made in advance, not a judgement made when the event was emitted. Even so, an allowance signals approval more explicitly than a received transfer, which could come from market making, wash trading (Cong et al., 2023), airdrop farming (Luo et al., 2025) or routing without any

intent to endorse. This holds only for a token contract that follows the standard: a spam or address-poisoning token can emit an `Approval` log that names an owner who never approved anything, which is one reason the study keeps only five widely traded tokens (Section 3.5.1).

Approvals have been read as a relation between addresses before: the self-authorization heuristic of Victor (2020) uses ERC-20 approvals to cluster addresses that it attributes to one entity. That reading links addresses and ranks nothing. It is also a caution, since an approval between two addresses of one entity is a self-endorsement, the edge that the Sybil cost model of Section 5.6 prices at gas alone.

2.8 What this research does not claim

This thesis ranks contracts by how other addresses relate to them on chain, and its claims stop at that ordering. No score is offered as a predictor of credit default or of any borrower's repayment, and no score is tested against loan defaults. No score is a security audit: a contract can contain malicious functions (Z. Lin et al., 2024) whatever its standing among owners, and the scores read neither code nor upgrade permissions. No score measures trading success. Where GMX V2 trader contracts enter the evaluation (Sections 3.7.1 and 3.7.2), their outcomes serve as out-of-window labels, not as a target that any score was built to predict. Agreement of a score with checks built from its own edges is an intended-construct check (Cronbach & Meehl, 1955), so part of it is built in, and the holdout coefficients are associations over time. None of the results supports a causal claim, for instance that approvals make a contract safer or that a high score brings a contract later approvals.

2.9 Rank correlation theory

Reputation scores and the proxies they are checked against are both continuous variables, and neither has a natural boundary for classification. Following the evaluation design of AWP (T. Do et al., 2023) and classical psychometrics (Cronbach & Meehl, 1955), this research measures two kinds

of rank agreement, monotonic and pairwise. Bootstrap confidence intervals quantify the sampling uncertainty of every coefficient of the main analysis (Efron, 1979).

Spearman's rank correlation (Spearman, 1904) is the Pearson correlation of the ranks. For paired ranks (R_i, S_i) of n items without ties, it reduces to

$$\rho = 1 - \frac{6 \sum_{i=1}^{n} (R_i - S_i)^2}{n(n^2 - 1)}. \tag{2.5}$$

This no-ties form does not hold when ties occur. Spearman's ρ is then the Pearson correlation of average ranks (mid-ranks), in which tied items share the mean of the positions they occupy, and every reported ρ is computed this way. In both forms ρ measures monotonic association.

Kendall's τ (Kendall, 1938) rests on counts of concordant and discordant pairs. Of the $n_0 = n(n-1)/2$ pairs (i, j) with $i < j$, n_c are concordant (both rankings order them the same way) and n_d are discordant (the rankings order them oppositely); a pair tied in either ranking is neither. Let $n_1 = \sum_k t_k(t_k - 1)/2$, where t_k is the size of the k-th group of tied values in the first ranking, and define n_2 in the same way for the second ranking. The τ-b variant is then

$$\tau_b = \frac{n_c - n_d}{\sqrt{(n_0 - n_1)(n_0 - n_2)}}, \tag{2.6}$$

and without the tie corrections it becomes $\tau_a = (n_c - n_d)/n_0$. Every τ reported in this thesis is τ_b, computed on the raw scores and proxies. The difference matters for heavily tied scores, such as the raw degrees, which are integer counts, and any PageRank that gives many contracts the same value (Section 4.4). For agreement about pairwise orderings, Kendall's τ is usually the easier coefficient to interpret, and Chapter 4 uses it as the main family-level summary. Its values are compared with those of other methods on the same cohort; no uncited absolute benchmark is used.

2.10 Theoretical framework

This study compares EndorseRank and AWP on reputation structure (H1), construct alignment (H2) and computational cost (H3), and then on prediction out of window (RQ4) and on coupling the two layers (RQ5). Three bodies of theory frame the first part: hyperlink–citation propagation, construct (domain) alignment, and computational complexity. Two further arguments bear on the second part.

- Hyperlink–citation semantics: In the original PageRank algorithm (Page et al., 1999), an inbound hyperlink counts as a citation. H.-D. Do and Do (2023) apply this reasoning to Ethereum transfers, and AWP (T. Do et al., 2023) weights each transfer edge by the age and the value of its transfers, through a logistic decay in age and a bounded transform of value (Section 3.6.1); this study sets the slope b of that transform from the median US-dollar amount (Section 3.5.1). EndorseRank extends citation-based propagation to ERC-20 approve and permit edges, in which the cited node is a spender contract and the citing nodes are the owners that approve it. An allowance is an explicit, on-chain delegation of spending authority that remains in force until it is revoked or replaced (Vogelsteller & Buterin, 2015), whereas a transfer may reflect trading or routing without any intent to trust. Risk propagation on Ethereum rates accounts that address clustering and illicit-account classification leave unassessed (D. Lin et al., 2025), so network structure can carry information that per-account methods miss, which argues for a directed endorsement graph alongside transfer volume.
- Domain alignment framework: Measurement theory uses construct alignment to ask how closely an indicator follows the construct it is meant to track (Cronbach & Meehl, 1955). Messick (1995) widens the idea into one unified judgment about what a score means and what using it implies. With rank-based reputation scores, Spearman's ρ and Kendall's τ measure monotonic and pairwise agreement in ordering and need no classification threshold, as in the evaluation of AWP (T. Do et al., 2023). Graph coherence is checked with contemporaneous transfer and allowance statistics of each contract. The out-of-window checks are temporal

holdouts: new approval pairs and new transfer senders for spender contracts, and outcome labels built from neither edge type for the contracts that trade on GMX V2, among them the liquidation-free close rate, whose definition the registration of W1 fixed (Sections 3.7.1 and 3.7.2).

- Computational complexity: Each PageRank iteration costs $O(|E|)$ (Section 2.5.2). Allowance graphs are sparser than transfer graphs, with fewer edges $|E|$, so an iteration needs less time and memory. The number of iterations is a separate factor, whose rate of convergence is governed by the damping factor (Langville & Meyer, 2006), so a sparser graph is not expected to need fewer iterations. This argument underlies H3. Both methods use the same solver, so differences in cost come from the graphs and from the restart vectors, which can change the number of iterations (Chapter 4 reports the edges and iterations of each solve); a speed that comes from a sparser graph does not count as an algorithmic contribution.
- Local counts as a baseline: Where contracts that already have many counterparties gain the most new ones (Section 2.4), the raw degree can already rank future arrivals, and a propagated score can beat it only when who the endorsers are matters more than how many. RQ4 therefore sets each PageRank against its own raw degree at the freeze date.
- Layer coupling: Multiplex PageRank (Halu et al., 2013) and multilayer ranking (De Domenico et al., 2015) treat several relations as layers over one node set. Mixing the layers node by node, as C-PR does, leaves a question open for layers of very different density: a node with no edge in the sparse layer follows the dense one, so the signal of the sparse layer may or may not survive. RQ5 therefore asks two things: whether the mixture improves on the transfer walk, and whether it beats both layers.

The conceptual model in Figure 2.3 links the scores of the thesis to H1–H3 and RQ1–RQ4. RQ5 is the dashed box on the right. It takes input from H2/RQ2 and from RQ4 and asks whether the two constructs combine.

Only EndorseRank and AWP enter H1. H2/RQ2 and H3/RQ3 concern every score, and RQ4 is

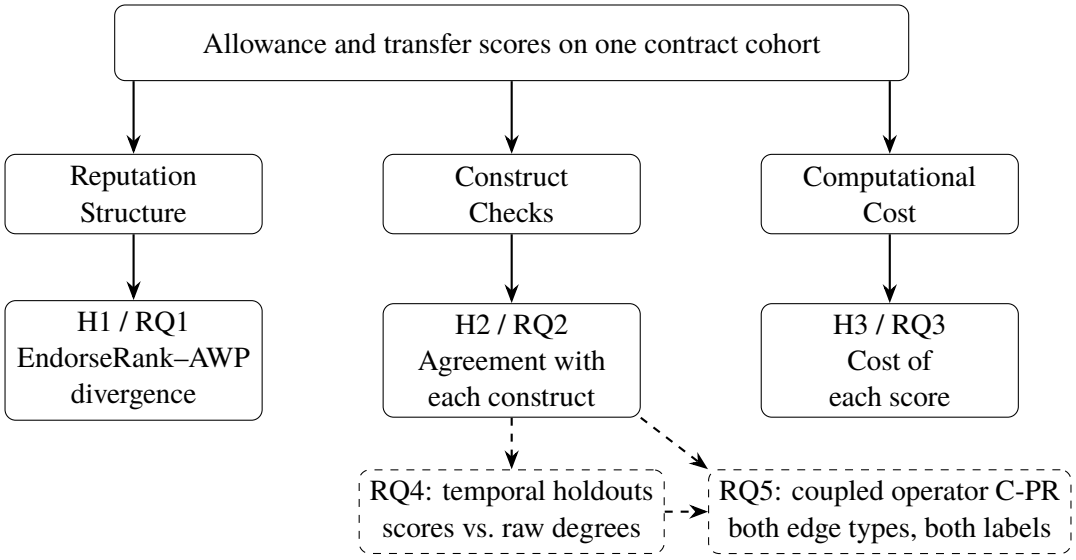


Figure 2.3: Conceptual model linking the allowance and transfer scores of one contract cohort to reputation structure (H1/RQ1), construct checks (H2/RQ2), computational cost (H3/RQ3) and the temporal holdout, in which each PageRank is set against the raw degree it smooths (RQ4). The dashed box on the right is RQ5, the coupled operator C-PR.

a time-split test on the spender contracts, with the raw degrees at the freeze as its baselines. The contracts that trade on GMX V2 are evaluated on outcomes built from neither edge type: profit labels in an exploratory W0 run, and the liquidation-free close rate in a comparison whose rules were registered before W1 was extracted and are applied to the revised scores after the registered results were known (Sections 3.7.1, 3.7.2 and 3.5.1).

2.11 Terminology

The formal constructs are defined in Chapter 3 (Sections 3.1 and 3.6). The terms below are used when results are interpreted:

- Single-layer methods: EndorseRank (allowance graph) and AWP (transfer graph), the two PageRank scores that walk on one edge type. The coupled walks of Section 3.6.3 and the raw degrees are evaluated beside them.
- Primary graph construct: the graph that a method is *designed* to rank, for example G_{approve} for

EndorseRank, as distinct from the checks applied after scoring.

- Contemporaneous check: a local transfer or allowance statistic taken from the same window as the score (Cronbach & Meehl, 1955).
- Temporal holdout: scores are frozen at t_1, and new owner–spender approval pairs and new transfer senders of the five tokens are counted from t_1 up to t_2 (Section 3.7); a second window, W1, registered before its labels were extracted, repeats the design with a later freeze and labels counted after the observation window ends (Section 3.7.2). Appendix 8.6.10 sets the decision contrasts of the registered analysis beside those of the revised one.
- Matched contract cohort, spender holdout cohort, trader cohort and benchmark subsamples: the address sets of Section 4.1.1. Every scored and evaluated address in them is a contract; EOAs enter the graphs only as owners, senders and recipients.

2.12 Domain alignment metrics

Construct alignment is measured by Spearman's ρ and Kendall's τ between scores and check rankings (Section 2.10), on contemporaneous transfer and allowance checks and on the holdout labels; Chapter 3 gives the operational definitions. Each coefficient of the main analysis is reported with a 95% bootstrap interval, and two scores are compared through the paired difference of their coefficients on the same resamples (Efron, 1979); the value-slope check and the sample-definition stages of Section 4.5 are reported without intervals.

2.13 Worked example: PageRank on a four-node endorsement graph

Take four addresses $\{A, B, C, D\}$ joined by the latest-allowance edges

$$A \to B\ (2), \quad A \to C\ (1), \quad B \to D\ (3), \quad C \to D\ (1),$$

with damping $d = 0.85$. One reading is that A is an owner's EOA, B and C are two vault contracts that A has approved, and D is a lending pool that both vaults approve in turn so that it can pull the tokens they supply. The out-strengths are $o_A = 3$, $o_B = 3$, $o_C = 1$ and $o_D = 0$ (the last is dangling). The edges $B \to D$ and $C \to D$ are the only out-edges of their nodes, so each is followed with probability one and its weight has no effect on the scores. Teleportation adds $(1 - d)/4 = 0.0375$ to every node at every step, the first one included. One power-iteration step from the uniform vector $\mathbf{r}^{(0)} = \mathbf{1}/4$ gives $\mathbf{r}^{(1)} = (0.0906,\ 0.2323,\ 0.1615,\ 0.5156)$ for (A, B, C, D). The mass already tilts toward D, since both B and C endorse D, while A splits its mass between B and C. The iteration converges to the stationary vector $\mathbf{r} = (0.1375,\ 0.2154,\ 0.1765,\ 0.4706)$, so D comes first, B and C follow, and A is last. No address endorses A. Apart from teleportation, its whole score comes from the dangling rule, which spreads D's mass evenly over the four nodes ($d\, r_D/4 = 0.1000$). The example shows that a node gains score by *receiving* endorsements from nodes that are endorsed in turn, which is the citation logic of PageRank (Page et al., 1999).

Suppose the same addresses were linked by transfers instead of allowances. The transfers would in general join other pairs, or the same pairs with other per-pair totals. That gives another transition matrix P, and possibly a different ranking of exactly the same nodes. Hypothesis H1 is this situation in miniature: EndorseRank and AWP need not agree.

2.14 DeFi risk taxonomy and where reputation fits

Werner et al. (2022) separate the technical security of DeFi protocols from their economic security, and L. Zhou et al. (2023) place the attacks and accidents seen in practice in one reference frame. Bugs in contract code are smart-contract risks, one of the several levels at which DeFi vulnerabilities arise (W. Li et al., 2022). Token contracts add risks of their own: customizations that break the ERC-20 standard put the DeFi applications that handle those tokens at financial risk, and a large-scale study of Ethereum token contracts finds such behaviors in most of the contracts it examines (Z. Li et al., 2026). Some contracts are hostile by design. In a contract-related rug pull the developers

abandon a token project and take investors' funds through malicious functions written into the contract, and code analysis can flag such functions (Z. Lin et al., 2024). On rollups, risk also arises between layers: a systematization of rollup security separates architectural risks, such as sequencing and data availability, from risks in how applications integrate with the rollup, such as cross-domain messaging (Efimov et al., 2026). A stale price, or a feed under someone's control, is an oracle risk (Angeris & Chitra, 2020). Market risks (Perez et al., 2021) cover volatility and liquidation cascades, including stress in stablecoin and funding markets (Klages-Mundt et al., 2020), and the market crash of March 2020 showed how these two layers can interact (Gudgeon, Perez, et al., 2020). Token-weighted voting belongs to governance risk: in the on-chain governance of Compound, Uniswap and ENS, a small number of addresses hold most of the voting power (Fritsch et al., 2024). We study a narrower layer, *counterparty* or *behavioral* risk, at the level of the contract: whether the approvals and transfers a contract has received mark it as a counterparty that owners trust to spend their tokens.

On-chain reputation scores cover only part of the behavioral risk layer and are no substitute for collateral, auditing, or oracle design. Their purpose is to compress a system's relational structure (who authorizes whom, who pays whom) into an ordinal measure that others can audit and reproduce from public data. Within behavioral risk, EndorseRank looks at authorization and AWP at payment flows. Neither tries to predict exploits, rug pulls, phishing or governance attacks.

Finally, fake activity makes behavioral risk harder to measure. Wash trading inflates volume on decentralized exchanges (Victor & Weintraud, 2021) and, on a larger scale, on unregulated centralized exchanges (Cong et al., 2023). Front-running and related forms of extractable value create transfers that follow ordering strategies instead of relationships (Daian et al., 2020); Qin et al. (2022) estimate how much of that value is captured on Ethereum. Scam tokens on Uniswap (Xia et al., 2021) and cross-exchange arbitrage trades (Makarov & Schoar, 2020) likewise add transactions without forming sustained relationships. The transfer graph is particularly sensitive to fake volume, while the allowance graph is vulnerable to approval farming and to approvals obtained by phishing (Section 2.17). Time-based splits, Sybil detection and a list of widely traded tokens, which keeps out

the logs of spam and address-poisoning tokens (Section 3.5.1), each solve the problem only in part.

2.15 Related measurement traditions

Outside blockchains, reputation systems have been studied in e-commerce (Resnick & Zeckhauser, 2002), in peer-to-peer networks (Kamvar et al., 2003), and in work on web spam (Gyöngyi et al., 2004). Four insights from this work apply here: (i) reputation is a property of relations; (ii) when identities are easy to create, Sybil attacks become possible (Douceur, 2002); (iii) seeding and normalization limit the effect of inflation; and (iv) evaluation needs external benchmarks, because internal coherence alone is not enough (Cronbach & Meehl, 1955). EndorseRank takes up (i) directly and treats (ii) as a threat. It does not take up (iii), since it restarts uniformly (Section 2.6). On (iv) it goes part of the way: the contemporaneous checks, measured by Spearman's ρ (Spearman, 1904) and Kendall's τ (Kendall, 1938), test coherence, while the temporal holdouts test prediction out of window, on the same constructs for spender contracts and on outcomes built from neither edge type for trader contracts.

Personalized PageRank (Jeh & Widom, 2003) and topic-sensitive PageRank (Haveliwala, 2003) bias the teleportation vector toward a chosen group of nodes, the mechanism that TrustRank uses to turn a small seed set into a global ranking (Gyöngyi et al., 2004); AWP's restarts in proportion to sending activity are another instance (T. Do et al., 2023). Multiplex PageRank lets a node's centrality on one layer shape the walk on another (Halu et al., 2013), and the multilayer framework of De Domenico et al. (2015) lets one walker move within and between the layers of an interconnected network. Both treat layers as different relations over one node set, which matches allowances and transfers among the same addresses. C-PR takes the simplest form of this idea, a per-node weighted sum of two transition matrices with one weight, and S-PR borrows the personalized-teleportation idea, with the allowance score as the teleportation vector (Section 3.6.3).

2.16 Accounts, gas, and why logs are the measurement surface

Ethereum has two kinds of account: externally owned accounts (EOAs) and contract accounts. A private key controls an EOA, whereas a contract account is governed by the code it stores (Antonopoulos & Wood, 2018). The difference decides what a score can attach to. An EOA acts when its key holder signs, while a contract acts only as its code allows when another account calls it, so the reputation of a contract belongs to its code and to whoever controls that code, not to a person. In the ERC-20 transfer network the two kinds of account also behave differently: transfers that involve contracts, especially transfers between two contracts, show heavier-tailed and less stable statistical patterns than transfers between EOAs (Mukhia et al., 2026). This thesis scores contracts only, and EOAs enter its graphs as owners, senders and recipients (Section 4.1.1).

Every state-changing transaction consumes gas. Logs come from a contract's `LOG` instructions, which run when its code emits an event (the Solidity `emit` statement); writing to storage produces no log. A reputation system that means to be auditable must take its graph from logs and not from off-chain databases, because logs are a permanent part of the blockchain history that any full node or index provider can recompute independently (Narayanan et al., 2016). Both graphs of this study come from logs; valuing their amounts in US dollars adds a public series of daily token prices kept off the chain, which Appendix 8.4.2 compares with Chainlink's on-chain price feeds.

Gas prices also shape the graphs. Cheaper gas on L2 networks densifies G_{transfer} more than G_{approve}, because transfers recur routinely while an approval is granted once so that later transfers can go through; Chapter 4 reports the resulting graph sizes.

2.17 Approvals as security-relevant events

Unlimited approvals (`type(uint256).max`) save the owner from approving again, at a cost in safety. If the approved contract is compromised, or was malicious from the start, the owner's whole balance of that token can be taken until the owner resets the approval. An allowance is therefore a capability and not a record: whoever holds it can move the owner's tokens until it is spent or

revoked (Vogelsteller & Buterin, 2015). EndorseRank interprets a token approval as endorsing one spender's capability to move those tokens. This is why an allowance does not mean the same thing as a transfer, even though both involve the movement of tokens. A contract could receive many transfers and never be granted permission to move funds on anyone else's behalf.

Wang et al. (2022) find that unlimited approvals make up the majority of ERC-20 approvals on Ethereum and quantify the risk they carry. Unlimited approvals appear as amounts of $2^{256} - 1 \approx 1.16 \times 10^{77}$ base units. EndorseRank's value weight, applied to US-dollar amounts, gives them the full weight of one, which a finite allowance nears only at a few times the median amount. The allowance layer of C-PR counts any allowance up to a cap in US dollars, so an unlimited approval weighs no more than a large finite one; in the registered analysis, whose layer summed raw base units, it took almost all of its owner's row whenever that owner also held finite ones (Section 3.5.1; Appendix 8.6.10). Appendix 8.4.2 reports how many owners hold both kinds.

The capability is what attackers seek. In transaction-based phishing, fake websites lead users to sign transactions that let scammers take their assets, and the first empirical study of the attack on Ethereum detected and reported more than 26,000 such websites in about eight months (He et al., 2023). A later study of payload-based phishing, in which malicious payloads manipulate the contract calls a user signs, sorts the tactics into four main categories and finds more than 130,000 phishing transactions on Ethereum over 300 days (Chen et al., 2025). Phishing of this kind is now sold as a service: operators build the toolkits, called wallet drainers, affiliates deploy the phishing websites, and the proceeds are split through profit-sharing contracts (He et al., 2025). Approval phishing also drives high-yield investment frauds that run across several blockchains, and an analysis of eighteen popular wallets finds that weaknesses in how wallet software presents and validates token approvals make these frauds easier (Adamczyk, 2025).

These attacks bear on a contract-level score in two ways. An approval obtained by deception emits the same `Approval` event as an informed one, so a spender that collects approvals by phishing gains endorsement edges just as an honest router does. EndorseRank cannot tell the two apart from the event; only who the approving owners are, and how strongly they are endorsed in turn, can separate

them. In the other direction, an owner asked to sign an approval has little on-chain evidence about the contract in front of it, and a score that summarizes which other owners have approved the same contract is one possible input to that decision. This thesis tests neither score as a phishing or drainer detector (Section 2.8).

2.18 AWP in depth: what the transfer-graph baseline measures

Adaptive Weighted PageRank (AWP), proposed by T. Do et al. (2023), ranks the addresses of a transaction graph. Each transaction adds to its edge a logistic function of its time multiplied by a bounded function of its value, so that no transaction adds more than one to the edge weight. The walk restarts at each address in proportion to its activeness, the sum over the addresses it has sent to of the time weight of its latest transaction to each. The paper then checks the rankings against inbound degree and inbound value, taken as proxies for the economic attention an address attracts. The form of the decay (Equation (3.3)) and the in-degree/in-value proxies in Chapters 3 and 4 are taken from the AWP paper. The paper gives no parameter values; this study fixes the decay parameters in its plan and sets the slope b of the value transform from the median US-dollar amount (Sections 3.6.1 and 3.5.1), and its in-value proxy sums inbound transfers in US dollars. As published, the approach is entirely on-chain and rank-based, and it needs no off-chain identity of any kind; applied to several tokens, as here, it also needs their prices, so that amounts in different tokens can be added. In this thesis AWP is applied to the same cohort contracts as EndorseRank, through the transfers of the five tokens that they send and receive.

EndorseRank keeps the same template, AWP's edge weights under PageRank plus a rank-correlation check, and changes what the edges mean. In Chapter 4 we test, on the same contracts, whether allowance edges give an ordering that differs from AWP's and agrees with the allowance checks, whether the cost of the solve follows the smaller edge set, and whether scores taken at t_1 rank the new approval pairs gained by t_2 better than the raw in-approve degree does.

2.19 Comparison of graph reputation methods

Table 2.1 compares EndorseRank with the other approaches covered in this chapter by edge semantics, temporal model and evaluation design.

As Table 2.1 indicates, EndorseRank's contribution is the use of the latest allowance as the edge of a reputation ranking of contracts, evaluated out of window; it is neither the ranking algorithm nor the reading of approvals as a link between addresses (Section 2.7). The two rows without edges judge different things: Human Passport judges the person behind an address, and CRPWarner judges the code of a contract. Neither ranks a contract by the relations that other addresses have formed with it. On the operator side, our contribution is to couple the two relations into one walk, with a seeded ablation to measure how much this coupling contributes. The coupling is tested out of window against both layers walked alone, under two rules that were fixed before the data they are judged on were extracted, rule A in the plan and rule B in the registration of W1, and are applied to the revised scores after the registered results were known (Sections 3.7, 3.7.2 and 3.5.1).

2.20 Summary of the reviewed literature

Table 2.2 collects in one place the sources underlying this study. Each row records a source's methodology, the strengths this thesis draws on, the limitations it tries to address, and how the source relates to this work. The research gap stated in Section 2.21 is grounded in the evidence collected in this table.

Table 2.1: Comparison of selected graph reputation methods (conceptual).

Method	Edge semantics	Temporal model	Typical evaluation
PageRank (web) (Page et al., 1999)	Hyperlinks	None / static snapshot	Retrieval quality
EigenTrust (Kamvar et al., 2003)	Local trust ratings	Iterative aggregation	P2P file authenticity
TrustRank (Gyöngyi et al., 2004)	Links from seed goods	Static / seeded	Spam demotion
PageRank/ HodgeRank on Ethereum (H.-D. Do & Do, 2023)	Token transfers	Static window	Not reproduced in this thesis
AWP (T. Do et al., 2023)	Transactions, bounded value weights	Logistic decay of edges and restarts	In-degree / in-value
RiskProp (D. Lin et al., 2025)	Transactions, with a de-anonymization score	Propagation of risk	Labeled illicit accounts
Multiplex (Halu et al., 2013) / multilayer PageRank (De Domenico et al., 2015)	Several relations over one node set	Static layers	Comparison with single-layer rankings; versatility of top-ranked nodes
Human Passport (n.d.)	No edges; weighted credentials about the address holder, on-chain and off-chain	Current credential set	Admission threshold; known human and Sybil addresses
CRPWarner (Z. Lin et al., 2024)	No edges; functions in a contract's code	Code at the time of analysis	Contracts tied to known rug pulls
EndorseRank (this study)	Latest allowances of five tokens in US dollars, AWP's weights	Logistic decay of each latest approval	Allowance checks; t_2 new approvals
C-PR / S-PR (this study)	Allowances mixed with, or seeding, transfers; US-dollar amounts, allowances capped	Allowance snapshot and logistic decay of transfers	Holdout labels; registered rules applied to the revised scores

Table 2.2: Summary of the reviewed literature: methodology, strengths, limitations, and relation to this study.

Source	Methodology	Strengths	Limitations	Relation to this study
Page et al. (1999); Langville and Meyer (2006)	Eigenvector ranking of a hyperlink graph, solved by power iteration	Interpretable propagation; $O(\|E\|)$ per iteration; damping yields a unique Perron vector	Edges can only mean hyperlinks; no notion of time	Same solver and complexity argument, used unchanged for both methods compared
Kamvar et al. (2003); Gyöngyi et al. (2004)	Aggregation of trust in P2P networks; propagation from seeds to demote spam	Normalization and seed sets curb inflation; the adversary is modeled explicitly	Depends on explicit ratings or curated seeds; not applied on-chain	Informs the Sybil-cost discussion and the uniform-teleportation baseline
H.-D. Do and Do (2023)	PageRank and HodgeRank applied to Ethereum transaction graphs as a measure of social credit	Treats transfer connectivity as relevant to credit; data are fully public	Transfers conflate payment with trust; allowance semantics absent	Prior transfer-graph study, kept separate from AWP in this thesis
T. Do et al. (2023), AWP	Transaction PageRank with bounded value weights, logistic recency and restarts weighted by sending activity; rank correlation with in-degree/in-value, presented as validation	Evaluates on ranks, so no default labels are needed; recency modeled explicitly	Proxies are computed from the same transfer data that forms the graph (mechanical alignment); dense graph; no parameter values reported	Baseline, used as published on the same contracts (Section 3.6.1); the decay parameters are set here and b from the median US-dollar amount, and EndorseRank takes the same edge weights with the median allowance

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Table 2.2 (continued)

Source	Methodology	Strengths	Limitations	Relation to this study
D. Lin et al. (2025), RiskProp	Network propagation of account risk, paired with a de-anonymization score	Rates every account, including those that clustering and classification leave unassessed; checked against labeled accounts	Checked against illicit-account labels; aimed at fraud, not trust or authorization	Shows that propagation over transactions adds to per-account methods; not reproduced here, since this study uses no fraud labels
Wu et al. (2022)	Phishing detection through network embedding of Ethereum transaction graphs	Embedding that uses the amounts and times of transactions; more effective than earlier embedding methods	Needs labeled phishing addresses; its labels mark phishing, not creditworthiness	Backs the premise that graph position is informative; not a competing method
D. Lin et al. (2024); Yan et al. (2023); C. Zhou et al. (2024); Q. Liu et al. (2025)	Account labelling with a graph convolutional network; scam detection with biased random walks; detection of airdrop hunters and Sybil addresses from transaction graphs	Show that account roles, scams and coordinated multi-address behavior leave traces in graph structure	Need labels; classify addresses and do not rank them	Back the premise that graph position is informative, and frame the Sybil threat to an allowance ranking
Hassija et al. (2020); Kotb (2023)	Blockchain-data credit scoring using prospect theory and machine learning	Aimed directly at lending decisions	Need labels or off-chain features; only partly interpretable	A contrasting class; EndorseRank remains within auditable graph propagation

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Table 2.2 (continued)

Source	Methodology	Strengths	Limitations	Relation to this study
Palaiokrassas et al. (2024); M et al. (2026)	Machine-learning prediction of liquidations from multichain DeFi features; a DeFi credit score that adds features from a weighted transaction graph	Show that on-chain behavior and graph position carry credit-relevant signal	Need liquidation or default labels; score borrowers	Support the use of graph position; this study ranks spender contracts, which do not borrow
Bellini et al. (2020); Almasoud et al. (2020)	Surveys of blockchain-based trust and reputation systems	Taxonomies of the design space; a review of thirty studies of reputation systems built on smart contracts	Reputation taken from users' ratings of services; authorization edges not considered; nothing measured on L2	None of the surveyed designs uses latest-allowance edges
Victor (2020); Victor and Weintraud (2021); Cong et al. (2023)	Heuristics for clustering addresses, one of which links addresses through ERC-20 approvals; detection of wash trading on DEXs and exchanges	Measure the share of on-chain activity that is artificial or multi-address	No reputation score proposed	Closest prior use of approvals, as a clustering link and not a ranking edge; show why transfer-based signals are open to inflation
He et al. (2023); Chen et al. (2025); He et al. (2025); Adamczyk (2025)	Measurement and detection of transaction-based, payload-based and approval phishing and of drainer-as-a-service operations	Large datasets of phishing websites, transactions and accounts; analysis of how wallets present approvals	Aimed at detecting attacks, not at ranking the contracts that hold approvals	Show why an approval is security-relevant and that an approval edge can be obtained by deception (Section 2.17)

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Table 2.2 (continued)

Source	Methodology	Strengths	Limitations	Relation to this study
Z. Lin et al. (2024); Z. Li et al. (2026); Tang and Shi (2024); Efimov et al. (2026)	Code analysis of rug-pull functions and of token contracts that break ERC-20; security analysis of contracts on Arbitrum and of rollup ecosystems	Judge a contract by its code and by the chain it runs on	Say nothing about how owners treat a contract	Contract-level risk work that a relational score complements and does not replace
Cronbach and Meehl (1955); Messick (1995)	Psychometric construct validity	Distinguish what a score is designed to measure from external criteria	Not specific to blockchains	Basis for treating the contemporaneous checks as construct checks and the temporal holdouts as out-of-window evidence
Human Passport (n.d.); Siddarth et al. (2020); Bordeianu and Popescu (2026)	Weighted credentials about an address holder, on-chain and off-chain, against a passing threshold; reviews of proof-of-personhood defenses	In deployed use; adds off-chain evidence against Sybil attacks	A gate and not a graded ordering; relations between addresses play no part; Sybil strategies keep changing	Shows how trust is established in practice, and what a contract-level score does not offer: no personhood guarantee
Efron (1979); DiCiccio and Efron (1996)	Bootstrap resampling and bootstrap-based confidence intervals	Uncertainty for rank statistics with no distributional assumption; paired contrasts	Units must be exchangeable	Source of the 95% intervals and $\Delta\tau$ contrasts reported in Chapter 4

2.21 Research gap

Five gaps emerge from Table 2.2 and Section 2.15. First, every on-chain reputation method discussed here (PageRank and HodgeRank (H.-D. Do & Do, 2023), AWP (T. Do et al., 2023), RiskProp (D. Lin et al., 2025), and the designs surveyed by Bellini et al. (2020)) builds its graph from transfers or explicit ratings, and the attestation tooling deployed in DeFi scores the holder of an address without any graph of relations between addresses (Human Passport, n.d.). None uses the ERC-20 allowance, the on-chain record of a deliberate delegation of spending power that every token following the ERC-20 standard carries. Approvals have served address clustering and security analysis (Sections 2.7 and 2.17), and contract risk has been assessed from code (Z. Lin et al., 2024), but in the work reviewed here approvals have not served as the edges of a reputation ranking, and contracts have not been ranked by the approvals they receive. It remains unknown whether a latest-allowance endorsement graph orders spender contracts differently from a transfer graph (O1 and RQ1). A second gap lies in evaluation. The transfer-graph baseline checks its scores against in-degree and in-value computed from the same transfer data used to build the graph, so some agreement is unavoidable; no report applies this procedure to an allowance construct and its transfer counterpart with confidence intervals and paired difference tests on the same set of contracts (O2 and RQ2). Third, a runtime says little about a method unless it is reported with the edge count, the memory use and the number of iterations under a fixed protocol; otherwise a difference in speed cannot be traced to sparsity as opposed to implementation (O3 and RQ3). The fourth gap is methodological and has to do with time. Agreement with graph statistics from the same window, the check used to evaluate AWP, cannot show that a score predicts later behavior; the out-of-window check used here is instead a temporal holdout, with a label for each construct counted after the freeze and the raw degree at the freeze as the baseline (O4 and RQ4). Finally, multiplex PageRank (Halu et al., 2013) and multilayer ranking (De Domenico et al., 2015) have been applied to social networks and to other networks with several kinds of relation, but not to two on-chain relations over one set of addresses. No study reports whether combining an authorization layer with a payment layer im-

proves on the payment layer alone, or beats both layers, under decision rules written down before the outcome is seen (O5 and RQ5). This study fixed such rules in its plan and in the registration of W1, and applies them to the revised scores as a re-analysis after the registered results were known (Section 3.5.1).

2.22 Open problems motivating this research

The literature leaves several problems in on-chain reputation unresolved, and these motivate the present study:

1. Edge semantics: When identities are pseudonymous (D. Lin et al., 2024), which on-chain relations capture trust or authorization best?
2. Evaluation without defaults: On what basis can methods be compared when credit records are missing (H.-D. Do & Do, 2023), and when the ranked contracts do not borrow at all?
3. Computational cost: As graphs grow, can reputation scores still be refreshed at interactive latencies (Langville & Meyer, 2006)?
4. Time-split checks: What keeps an evaluation from mistaking same-window graph statistics for out-of-window evidence (Cronbach & Meehl, 1955)?
5. Adversarial robustness: How much does it cost to forge reputation when owner addresses and contracts are cheap to create (Douceur, 2002)?

This study takes on (1)–(4) through allowance edges, contemporaneous checks, runtime benchmarks and temporal holdouts. Problem (5) is analyzed at the model level in Chapter 5; a full empirical adversarial evaluation is left for future work.

Chapter 3 turns this gap statement into an operational design: the data window and cohorts, the construction of both graphs and their solution with one shared PageRank implementation, the proxy families, the temporal holdout, and the timing and bootstrap protocols behind the results of Chapter 4.